

Piyush Marmat

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Education

Doctor of Philosophy - Astronomy

Mar 2026 – Sep 2028

Janusz Gil Institute of Astronomy, University of Zielona Góra, Poland

Advisor: dr hab. Wojciech Lewandowski, prof. UZ. dir. JGIA.

Thesis: *Probing Anisotropy and Inhomogeneity in Interstellar Medium with Long Term Radio Pulsar Observations*

Integrated Master of Science in Physics (Dual-Degree)

Jul 2017 – May 2022

Indian Institute of Technology Roorkee, Roorkee, Uttarakhand, India

Advisor: Prof. Paramasivan Arumugam, Department of Physics, IIT Roorkee.

Thesis: *Equation of State based on phenomenological relativistic mean field theory for single and double fluid Neutron Stars*

Publications

1. Planets in Pulsar Winds. T Kaister, S Méndez, P Marmat, M Čemeljić, M Velli, J Varela, M Falanga. arXiv preprint arXiv:2605.14077, 2026. <https://doi.org/10.48550/arXiv.2605.14077>.
2. Jaikhomba Singha, Bhal Chandra Joshi,..., Piyush Marmat et al. Improving DM estimates using low-frequency scatter-broadening estimates, Monthly Notices of the Royal Astronomical Society, Volume 535, Issue 1, November 2024, Pages 1184–1192, <https://doi.org/10.1093/mnras/stae2405>.
3. Branchesi, M., Falanga, M., Harms, J.,..., Piyush Marmat et al. Lunar Gravitational-Wave Detection. Space Sci Rev 219, 67 (2023). <https://doi.org/10.1007/s11214-023-01015-4>.
4. Joshi, B.C., Gopakumar, A., Pandian A.,..., Piyush Marmat et al. Nanohertz gravitational wave astronomy during SKA era: An InPTA perspective. J Astrophys Astron 43, 98 (2022). <https://doi.org/10.1007/s12036-022-09869-w>.
5. Tarafdar, P., Nobleson, K., Rana, P., Singha, J., Krishnakumar, M., Joshi, B., . . ., Piyush Marmat, . . . Pathak, D. (2022). The Indian Pulsar Timing Array: First data release. Publications of the Astronomical Society of Australia, 39, E053. <https://doi.org/10.1017/pasa.2022.46>.
6. K Nobleson, Nikita Agarwal,. . ., Piyush Marmat et. al. Low-frequency wideband timing of InPTA pulsars observed with the uGMRT, Monthly Notices of the Royal Astronomical Society, Volume 512, Issue 1, May 2022, Pages 1234–1243, <https://doi.org/10.1093/mnras/stac532>.
7. Jaikhomba Singha, Mayuresh P Surnis,..., Piyush Marmat et. al. Evidence for profile changes in PSR J1713+0747 using the uGMRT, Monthly Notices of the Royal Astronomical Society: Letters, Volume 507, Issue 1, October 2021, Pages L57–L61, <https://doi.org/10.1093/mnrasl/slab098>.

Conference Contributions

1. *How magnetospheric propagation affects the polarised radio emission from pulsars*, Conference Presentation, 27th RAGTIME, Silesian University in Opava, Czech Republic, Nov 2025.
2. *Testing Polarization Models on the Radio Pulsar PSR J1900-2600: Birefringent Filtering, Angular Mode Separation, and Mode Coupling*, Conference Presentation, 42nd Congress of the Polish Astronomical Society, University of Warsaw, Poland, Sep 2025.
3. The Glitch Rise Times of Rotating Neutron Stars. J Singha, H Grover, P Marmat, P Arumugam, Proceedings of the DAE Symp. on Nucl. Phys 66, 798. (2022)
4. Neutron Star Equation of state in single and two fluid formalism. B Bhat, P Marmat, H Grover, J Singha, A Kumar, P Arumugam, Proceedings of the DAE-BRNS symposium on nuclear physics. V. 66, 2-2. (2022)
5. A python based pipeline to analyse Interstellar scintillation of radio pulsars observed with ORT and uGMRT. P Marmat, J Singha, BC Joshi, Poster - 40th Meeting of ASI 2022.
6. Building a 5m aperture small radio-telescope at IIT Roorkee. Vanshaj Kerni, Piyush Marmat et. al. Conference Poster, 40th Meeting of ASI 2022.
7. Using low frequency scatter-broadening measurements for precision estimates of DM. Jaikhomba Singha, Bhal Chandra Joshi, P. Arumugam, Piyush Marmat, et. al. Conference Poster, 40th Meeting of ASI 2022.

Professional and Scientific Experience

Visiting Researcher, Nicolaus Copernicus Astronomical Center, Toruń, Poland.

Mar 2025 - Feb 2026

- Created an open-source web application for analysis of polarisation properties of radio pulsars using MeerKAT single-pulse data (with dr habil. Jarosław Dyks).

- Visited Prof. Harish Vedantham (ASTRON, The Netherlands) to collaborate on the observability of coherent radio emission from pulsar planets, supervised by Dr. Miljenko Čemeljić.

Visiting Scientist, International Space Science Institute, Bern, Switzerland.

Mar-Apr 2023

- Contributed to a white paper on lunar gravitational-wave detection, including analysis of tidal disruption of white dwarfs.
- Improved Python simulations for phase-resolved spectra and frequency-resolved light curves of magnetic cataclysmic variables in the X-ray regime, including interstellar photoelectric absorption.
- Analyzed the Gertsenshtein-Zel'dovich effect in the context of pulsar magnetospheres.
- Assisted late Prof. Dr. Maurizio Falanga in graduate-level teaching activities and grant proposal preparation.

ThinkSwiss Research Scholar, University of Bern, Switzerland.

Sep-Nov 2022

- Implemented radial velocity models for spectroscopic binaries; simulated dynamics and velocity profiles.
- Derived fractional changes to pulsar spin induced by gravitational waves leading to quadrupolar oscillations.
- Compared the effect of magnetospheric radii on accretion luminosities in neutron stars and rotating black-holes.

Associate Research Fellow, Indian Pulsar Timing Array Consortium, India.

May 2021 - Dec 2022

- Co-authored five peer-reviewed publications with contributions to uGMRT pulsar observations, data reduction, Cyber infrastructure management and web development.
- Simulated time and frequency evolution of dispersion measure and scattering parameters.

Remote Intern, National Centre for Radio Astrophysics, Pune, India.

Apr 2020 - Mar 2021

- Developed a Python and PSRCHIVE based script to analyze scintillation spectra of pulsars observed with ORT and uGMRT.

Head Secretary, Physics and Astronomy Club-IIT Roorkee, India.

Jul 2019 - Apr 2021

- Served as the joint and then head secretary leading a team of 40+ undergrad students, supervising several learning projects, organizing several events including seminars by distinguished scientists worldwide.
- Led a team of 3, secured 1st position nationally; in IIT Kanpur Hyperion Case Study Challenge (Oct 2021), involving galactic dark-matter density modeling.
- With a team of 8 and won a silver medal representing the institution at national level at the 9th Inter-IIT Tech-Meet, IIT Guwahati by developing a web-based catalog for observations made through the ISRO's AstroSat (Mar 2021)

Training / Workshops / Schools

- REBOUND N-body simulation workshop at the Institute of Astronomy, University of Zielona Góra, Mar 2026.
- 25th Relativistic Astrophysics Group Meeting, Silesian University in Opava, Czech Republic Nov 2023
- Forum on Lunar Gravitational-Wave Observation, ISSI-Bern, Switzerland, Oct 2022
- HPC Workshop on Radio Astronomy Data Analysis in the SKA Era, IIT Roorkee, India, Mar 2022.
- 40th Meeting of the Astronomical Society of India, IIT-Roorkee, India, Mar 2022.
- Second Chennai Symposium on Gravitation and Cosmology - Online, Feb 2022.
- Online IR Astronomy School organised by SOFIA-NASA, Feb 2022
- TIFR-ICTS CSGP Online Meeting, Dec 2021.
- GBO and Arecibo Online Single Dish School, September 2021.
- IPTA Online Student Week, Jun 2021.

Scientific Outreach

- Delivered an invited online colloquium on career in space science at the St. Paul's Elementary School in Akola, Maharashtra in India on 23rd of August 2025 (Indian Space Day).
- Presented my works on the interactions of high-frequency gravitational waves with neutron stars and pulsars in a student-colloquium organized by Prof. Martin Urbanec at the Silesian University of Opava, Czech Republic, Dec 2023.
- Hosted a workshop on the basics of observational astronomy and using the Stellarium open-source planetarium software, organized by the Student Technical Council of IIT Roorkee, Dec 2020.
- Presented a brief review of the Betelgeuse's dimming in late 2019 which suspected its untimely supernova explosion in a student-colloquium organized by Indian Physics Association-Roorkee Chapter, Feb 2020.